Fiscal Year:	FY 2008	**Task Last Updated:**	FY 03/03/2008
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Project Title:	High Pressure Studies of Carbon Monoxide Oxidation for Application to Mars Exploration		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	11/18/2003	**End Date:**	11/17/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/17/2007 per PI (4/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GA20G		
Performance Goal No.:			
Performance Goal Text:			
	The investigation of the high pressure combustion chemistry of carbon monoxide was proposed for application to the rocket engine propulsion of a Mars surface vehicle. The rocket engine will be fueled with carbon monoxide and oxygen produced in-situ from the electrolysis of Martian atmospheric carbon dioxide. In-situ production has the advantage of eliminating the need to transport fuel and oxidizer from Earth but has, in the case of a carbon monoxide/oxygen rocket engine, the disadvantage of generating a fuel with relatively low performance in the rocket engine. Consequently, the maximum performance possible from a carbon monoxide engine is necessary and optimal engine design is required to achieve it. Part of the engine design involves an evaluation of the effect of finite rate oxidation chemistry of carbon monoxide on performance. The evaluation of the chemistry necessitates a detailed chemical kinetic model that for the best engine design needs to capture very accurately all the chemical steps that influence the combustion of carbon		

Task Description:	monoxide. There are detailed chemical kinetic models available that do include the most current understanding of carbon monoxide oxidation but they have never been validated by experiments conducted at the operating pressures of the proposed Martian vehicle rocket engines: 200-300 psi (13.7-20.4 atm) for pressure fed engines, 2000-3000 psi (136-204 atm) for turbopump fed engines. These high pressures are rarely achievable in laboratory experiments that investigate chemical kinetics but are routinely accessed in the very high pressure single pulse shock tube operating at the University of Illinois at Chicago. This shock tube can create pressure ranges from 5-1000 atm, temperatures currently from 950-1450 K but extendable to 1900K and reaction times from 0.5-2.5 msec. These are the ranges of conditions that will used to experimentally examine the bone dry and hydrogen containing oxidation of carbon monoxide. Gas chromatography/mass spectrometry/thermal conductivity measurements of the carbon monoxide, oxygen, carbon dioxide, water species concentrations as a function of pressure, temperature, hydrogen content or water content, and stoichiometry can be compared with the predictions from existing models. Prior work examining the comparison between predictions of existing models for hydrocarbon oxidation and high pressure results obtained from the shock tube have shown that hydrocarbon models developed and validated against lower pressure experiments require modification in order to predict high pressure results. Therefore the hypothesis underlying the experimental examination of the oxidation of carbon monoxide is that current chemical kinetic models for this reactant system need to be modified to account for high pressure effects. The modifications of these models will lead to a more accurate, comprehensive detailed chemical kinetic model that when coupled with computer design codes currently used by NASA will lead to the highest performance possible in the Mars vehicle rocket engine. The task objective has been to use the unique single pulse shock tube at UIC to examine the oxidation of carbon monoxide with and without a hydrogen source over an extreme range of pressures and temperatures that encompass the design range of the Mars vehicle rocket engine so that recent models for carbon monoxide oxidation can be validated and made more comprehensive for application to the design of the high pressure rocket engine. To achieve the maximum performance from the Mars exploration vehicle rocket engine, which would be either pressure fed at 200-300 psi or pump fed at 2000-3000 psi, predictions or measurements of critical performance characteristics resulting from the use of carbon monoxide as the fuel are necessary. Among the critical characteristics are ignition of the combustible mixture, flame front heat release, and the effects of finite rate chemistry on specific impulse. These characteristics can be better predicted through an improved knowledge of the high pressure oxidation kinetics of carbon monoxide, as will be obtained in this task, and the better predictions necessary to achieve improved Mars rocket engine performance constitute the significance of the work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Carbon monoxide oxidation kinetics are an essential part of the modeling of the combustion of earth based hydrocarbon fuels such as gasoline, kerosene and jet propulsion fuels. Cleaner and more efficient combustion of these earth based fuels will come from better designs of combustion devices which have incorporated accurate models of hydrocarbon combustion into the design process. Therefore, an improved high pressure carbon monoxide oxidation model, that has come from this task work, could aid in the modeling and design of earth based combustion devices especially those operating at high pressures.
Task Progress:	CO oxidation experiments were performed over a wide temperature range (1000-1500 K) for stoichiometric (phi=1) and fuel lean (phi=0.5) mixtures. Seven sets of data were obtained with abundant amount of H2 in the reactant mixture, four data sets at nominal pressures of 24, 43, 256 and 450 atm for stoichiometric oxidation, three data sets at nominal pressures of 24, 43 and 256 atm for fuel lean oxidation, comprising more than 150 experiments. For all the experiments, profiles for CO, CO2 and O2 were obtained. No significant difference has been observed in the species profiles from the stoichiometric to the fuel lean mixtures for the 24, 43, and 256 atm data sets. On the other hand, the system shows pressure dependence as the pressure is raised from 24 to 256 atm and a slower decay for CO is observed. No significant difference has been observed in the species profiles as the pressure is further raised from 256 to 450 atm, indicating that reactions have reached the high pressure limit at 256 atm. The experimental data have been simulated using the most recent model for H2/CO combustion developed by Davis and co-workers, and GRI-Mech 3.0. While GRI-Mech is not able to predict correctly the species behavior, trends for CO, O2 and CO2 are well described by the Davis et al. model. However especially for the higher pressure data sets the model does not do a good job, and further work was required to improve the model. In particular sensitivity analysis and reaction pathway analysis showed the importance of the reactions H+O2=O+OH, H+O2+M=HO2+M, and CO+OH=CO2+H under all pressure conditions. The rate coefficients of these reactions have been extensively studied. However within their remaining uncertainties there was no possibility to significantly improve the accuracy of the simulation results over the entire pressure range. Because of the elevated pressures in the present study, reactions involving the HO2 radical start to become the next most influential reactions. Among these reactions, the reaction HO2+OH=O2+H2O has been extensively studied. In particular a new rate coefficient expression has been employed to take into account the very strong and peculiar temperature dependence of this reaction. With this single rate revision, the revised Davis et al. model can accurately simulate the shock tube data for the entire pressure range of 24-450 atm. The revision does not affect the accuracy of the model against the data sets presented in previous publications. An additional data set was obtained by decreasing the amount of H2 in the reactant mixture. The mixture used for the experiments was a fuel lean mixture composed of 235 ppm of CO, 48 ppm of H2 and 273 ppm of O2 in Argon. The ratio between the amount of H2 and CO (~ 0.21) is much smaller than the ratios of the mixtures previously studied (~ 0.36-0.40). The pressure at which the new data set was obtained was 256 atm, while the temperature varied from 1100 to 1550 K. The experimental procedure was slightly modified, as the analyses of the reactant and product species were conducted using an additional detector, the PDD detector. This detector is able to measure the mole fraction of H2 present in a mixture down to ppb levels. Such measurement is very important in order to test the accuracy of the revised Davis et al. model with respect to the amount of H2 present in the mixture.

The experimental results for CO and H2, have been compared with the simulations obtained using the revised Davis et al. model. The model does an excellent job in simulating both the CO and the H2 species profiles.

For a simple system as the one studied, the accuracy in simulating CO, CO2, O2, and H2 profiles implies accuracy also in predicting the amounts of all the other species present in the mixture, i.e. H2O. This feature makes the revised Davis et al. model an excellent tool for simulations and predictions over a wide range of pressures (up to 450 atm), temperatures (up to 1550 K) and initial mixture composition.

The model has been subsequently used for some complementary modeling studies. The goal of these studies is to understand how the kinetics of the carbon monoxide oxidation is influenced by the amount of hydrogen source present in the initial mixture.

CO profiles have been obtained for fuel lean (phi=0.5) mixtures at 2 different pressures (48 atm and 338 atm) and over a wide range of temperature (1150 K to 1600 K). The initial CO mole fraction is 225 ppm, while the amount of H2 is varied (25 ppm, 5 ppm, and 1 ppm). Thus the ratio between the mole fractions of H2 and CO is respectively 0.11, 0.02, and 0.004. These ratios are much smaller than the ones of the experimental mixtures; the reduction in the relative amount of H2 leads to a shift of the profiles toward higher temperatures. This shift affects the pressure dependence of the global reaction system; at low H2 mole fractions an increase in pressure leads to higher conversion of CO, in contrast to prior experimental results. This difference can be explained by investigating the relative importance of the single reactions at different temperature and pressure conditions. At all the conditions, the dominant reaction for CO consumption is CO+OH=CO2+O. Thus the rate of CO decay depends on the concentration of OH; the latter species is mainly formed through the reaction H+O2=OH+O. This is not the only possible channel for the recombination of H and O2 as in presence of a third body the two species can recombine forming HO2, i.e. H+O2+M=HO2+M. This termolecular reaction depends on both temperature and pressure; at relatively low temperatures, HO2 production increases with pressure and the amount of OH available in the system decreases. As a consequence, the CO conversion decreases, as observed previously. But as the system reaches higher temperatures, the backward rate of the reaction H+O2+M=HO2+M becomes predominant in comparison with the forward rate. The HO2 converts back to H and O2, forcing the production of OH through the reaction H+O2=OH+O. As a consequence, the CO conversion increases with pressure.

Similar trends are shown for mixtures containing H2O as hydrogen source. The type of hydrogen source, i.e. H2 or H2O, does not affect the dependence of the system on the pressure, but H2O has been found to be less effective than H2 in decreasing the overall activation energy.

Reference: S. G. Davis, A. V. Joshi, H. Wang, F. Egolfopoulos, Proc. Combust. Inst. 30 (2004) 1283-1292.

Bibliography Type:	Description: (Last Updated: 02/21/2014)
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